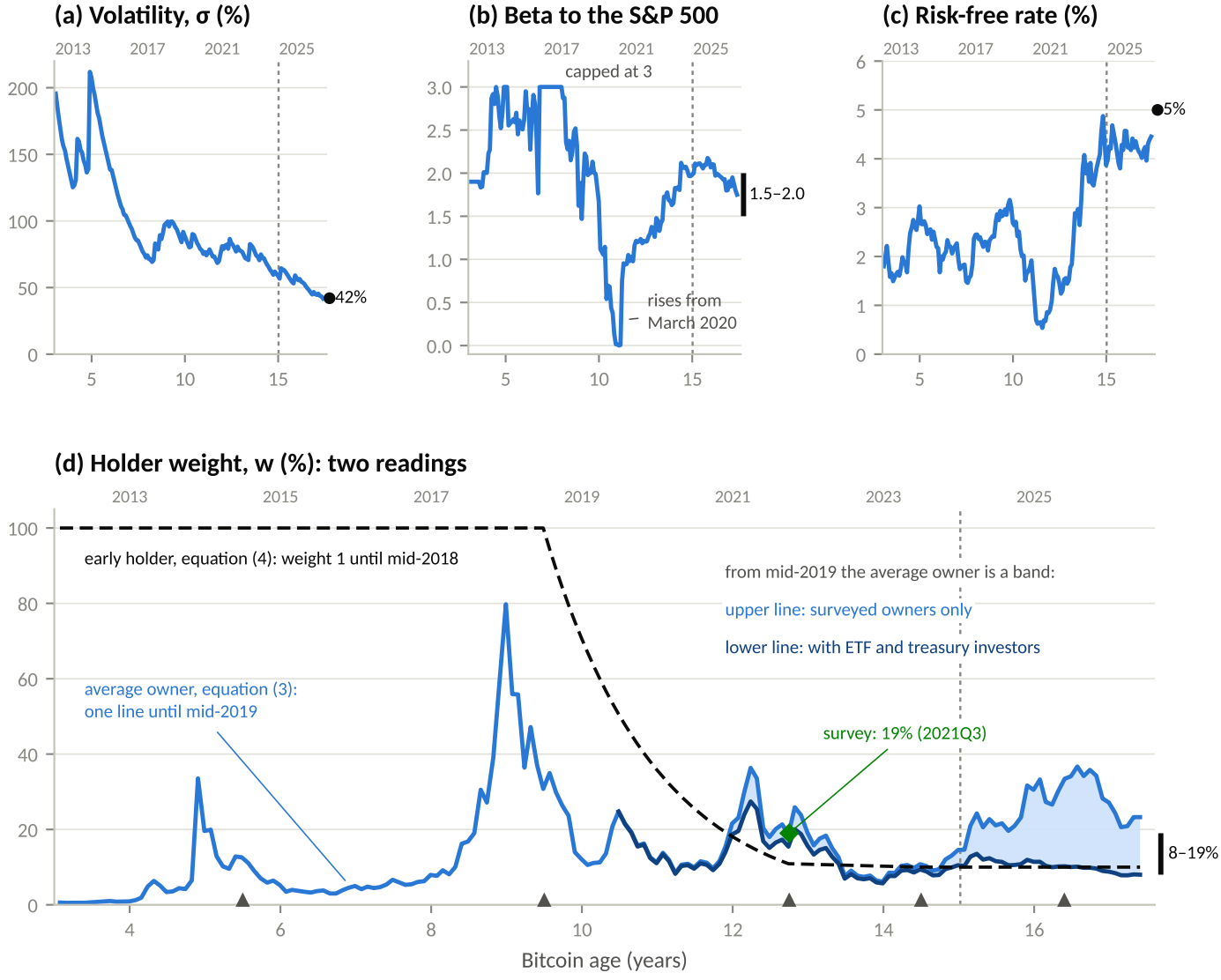

(a) Volatility, σ (%)
2013
2017
2021
2025
200
150
100
50
0
42%
5
10
15
(b) Beta to the S&P 500
capped at 3
3.0
2.5
2.0
1.5
1.0
0.5
0.0
1.5–2.0
rises from
March 2020
(c) Risk-free rate (%)
6
5
4
3
2
1
0
5%
(d) Holder weight, w (%): two readings
2013
2015
2017
2019
2021
2023
2025
100
80
60
40
20
0
early holder, equation (4): weight 1 until mid-2018
from mid-2019 the average owner is a band:
upper line: surveyed owners only
lower line: with ETF and treasury investors
average owner, equation (3):
one line until mid-2019
survey: 19% (2021Q3)
8–19%
4
6
8
10
12
14
16
Bitcoin age (years)